

Codificación de fuentes de información

Teoría de la Información y la Comunicación



Codificación de fuentes de información

C A D B C C B A C A B C C D B

A

3

B

4

C

6

D

2

Total

15

Codificación de fuentes de información

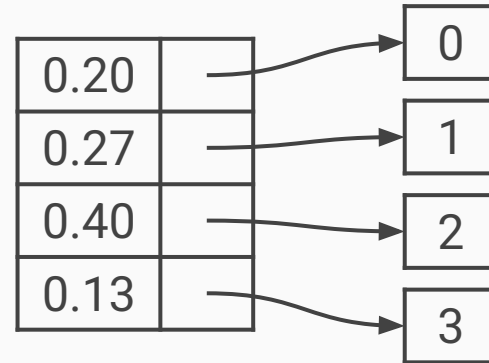
C A D B C C B A C A B C C D B

A	B	C	D	Total
0.20	0.27	0.40	0.13	1

Algoritmo de Huffman

A	0.20
B	0.27
C	0.40
D	0.13

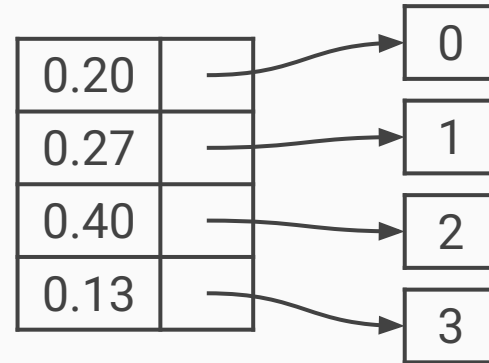
0	A	
1	B	
2	C	
3	D	



Algoritmo de Huffman

C	0.40
B	0.27
A	0.20
D	0.13

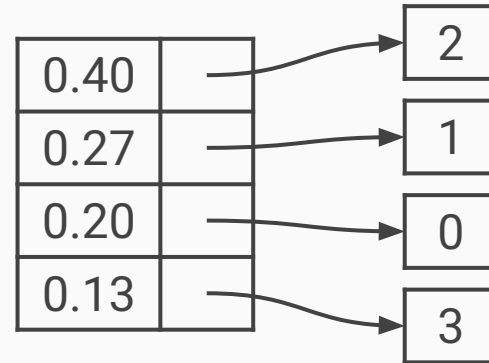
0	A	
1	B	
2	C	
3	D	



Algoritmo de Huffman

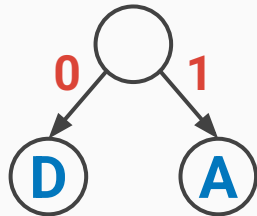
C	0.40
B	0.27
A	0.20
D	0.13

0	A	
1	B	
2	C	
3	D	

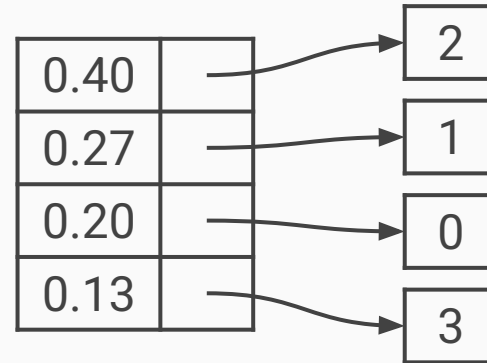


Algoritmo de Huffman

C	0.40
B	0.27
A	0.33
D	

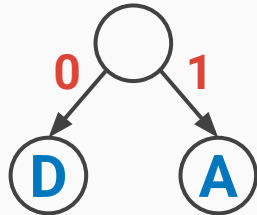


0	A	
1	B	
2	C	
3	D	

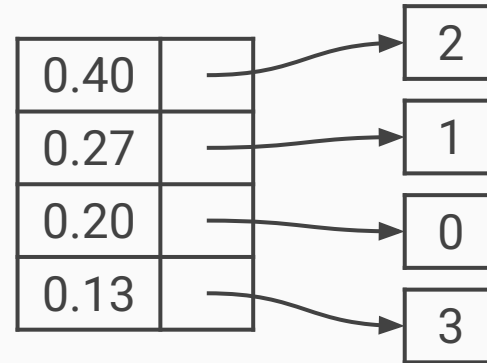


Algoritmo de Huffman

C	0.40
B	0.27
A	0.33
D	

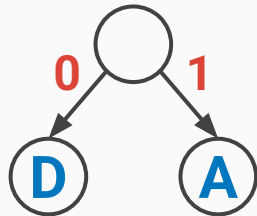


0	A	1
1	B	
2	C	
3	D	0

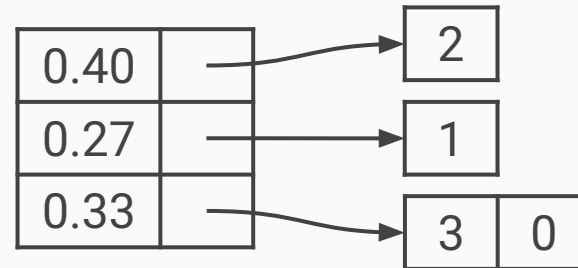


Algoritmo de Huffman

C	0.40
B	0.27
A	0.33
D	

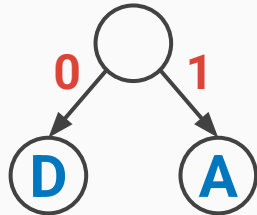


0	A	1
1	B	
2	C	
3	D	0

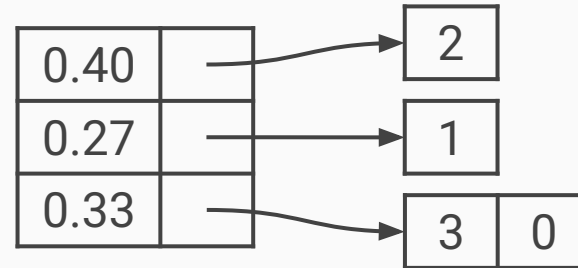


Algoritmo de Huffman

C	0.40
A	0.33
D	
B	0.27

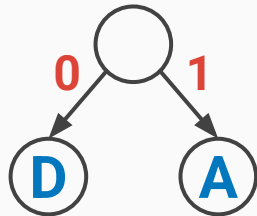


0	A	1
1	B	
2	C	
3	D	0

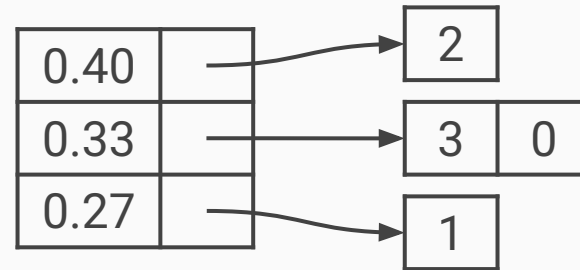


Algoritmo de Huffman

C	0.40
A	0.33
D	
B	0.27

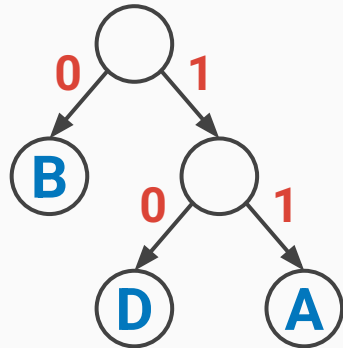


0	A	1
1	B	
2	C	
3	D	0

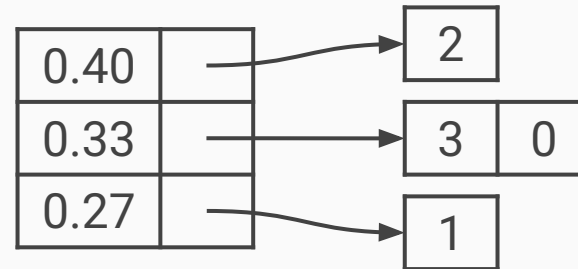


Algoritmo de Huffman

C	0.40
A	0.60
D	
B	

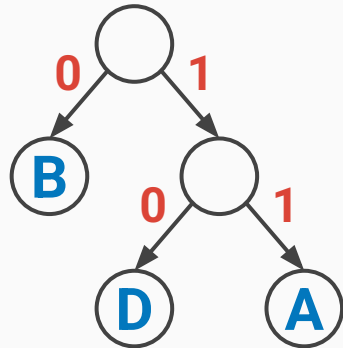


0	A	1
1	B	
2	C	
3	D	0

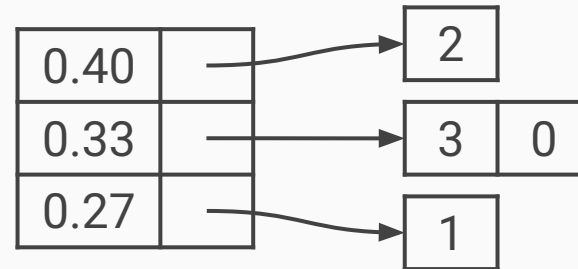


Algoritmo de Huffman

C	0.40
A	0.60
D	
B	

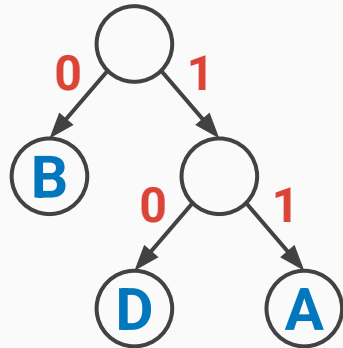


0	A	11
1	B	0
2	C	
3	D	10

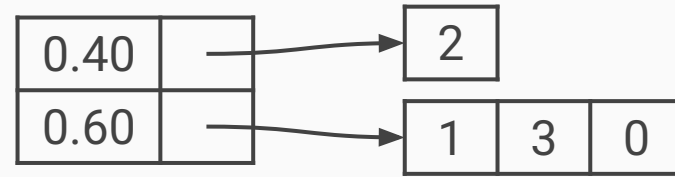


Algoritmo de Huffman

C	0.40
A	0.60
D	
B	

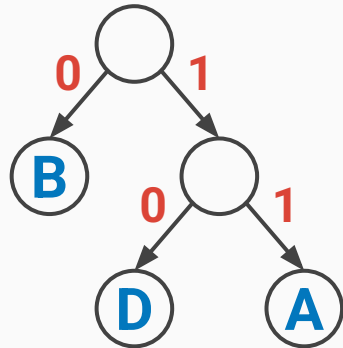


0	A	11
1	B	0
2	C	
3	D	10

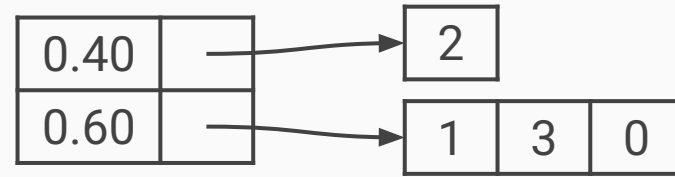


Algoritmo de Huffman

A	0.60
D	
B	
C	0.40

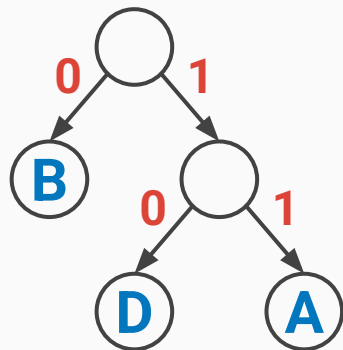


0	A	11
1	B	0
2	C	
3	D	10

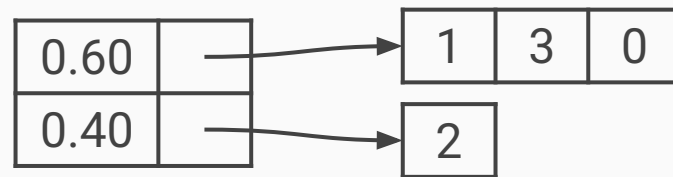


Algoritmo de Huffman

A	0.60
D	
B	
C	0.40

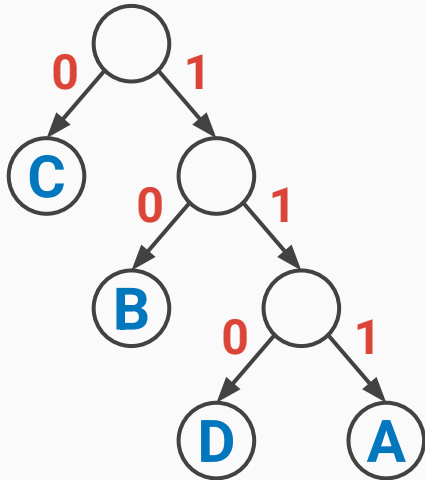


0	A	11
1	B	0
2	C	
3	D	10

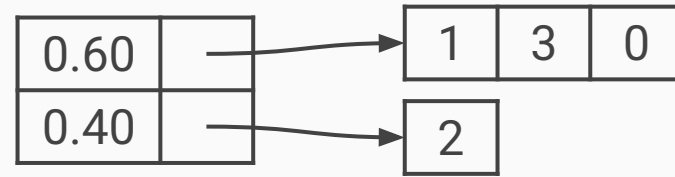


Algoritmo de Huffman

A	1.00
D	
B	
C	

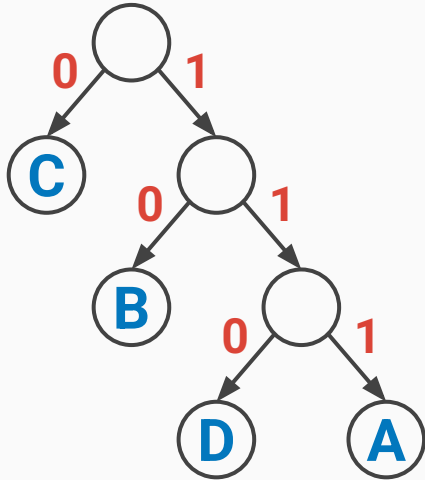


0	A	11
1	B	0
2	C	
3	D	10

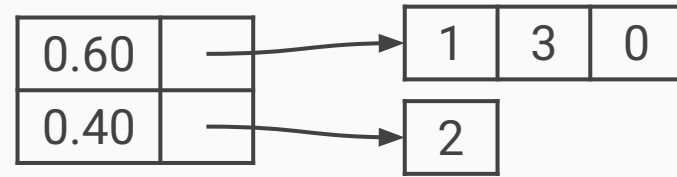


Algoritmo de Huffman

A	1.00
D	
B	
C	

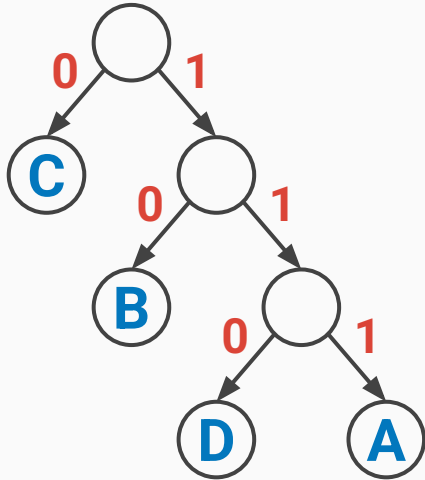


0	A	111
1	B	10
2	C	0
3	D	110

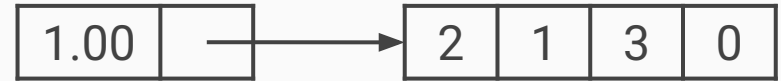


Algoritmo de Huffman

A	1.00
D	
B	
C	



0	A	111
1	B	10
2	C	0
3	D	110



Algoritmo de Huffman

A	0.20	111
B	0.27	10
C	0.40	0
D	0.13	110

$$H(S) = 1.89 \text{ bits}$$

$$L = 1.93 \text{ bits}$$

$$\eta = 0.9772$$

$$R = 0.0228$$

Algoritmo de Huffman

C A D B C C B A C A B C C D B

0 111 110 10 0 0 10 111 0 111 10 0 0 110 10

A	0.20	111
B	0.27	10
C	0.40	0
D	0.13	110

Algoritmo de Shannon-Fano

A	0.20
B	0.27
C	0.40
D	0.13

0	A	
1	B	
2	C	
3	D	

0.20	0
0.27	1
0.40	2
0.13	3

Algoritmo de Shannon-Fano

C	0.40
B	0.27
A	0.20
D	0.13

0	A	
1	B	
2	C	
3	D	

0.20	0
0.27	1
0.40	2
0.13	3

Algoritmo de Shannon-Fano

C	0.40	} 0.40
B	0.27	
A	0.20	} 0.33
D	0.13	

0	A	
1	B	
2	C	
3	D	

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	}	0.40
B	0.27		
A	0.20	}	0.60
D	0.13		

0	A	
1	B	
2	C	
3	D	

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	0
A	0.20	
D	0.13	

0	A	
1	B	
2	C	
3	D	

0.40	2	} 0.40
0.27	1	
0.20	0	} 0.60
0.13	3	

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	0
A	0.20	
D	0.13	

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2	} 0.40
0.27	1	
0.20	0	} 0.60
0.13	3	

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	0
A	0.20	
D	0.13	

} 0.60

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	0
A	0.20	
D	0.13	

} 0.27

} 0.13

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1
B	0.27	0
A	0.20	
D	0.13	

0.27

0.33

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2	} 0.40
0.27	1	
0.20	0	} 0.60
0.13	3	

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

0	A	0
1	B	0
2	C	1
3	D	0

0.40	2	}	0.27
0.27	1		
0.20	0	}	0.33
0.13	3		

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2	}	0.27
0.27	1		
0.20	0	}	0.33
0.13	3		

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

} 0.33

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

← 0.00

} 0.13

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1	
B	0.27	0	1
A	0.20		0
D	0.13		

} 0.20

} 0.13

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2	}	0.27
0.27	1		
0.20	0	}	0.33
0.13	3		

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	00
1	B	01
2	C	1
3	D	00

0.40	2	
0.27	1	
0.20	0	} 0.20
0.13	3	

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	001
1	B	01
2	C	1
3	D	000

0.40	2	
0.27	1	
0.20	0	} 0.20
0.13	3	

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	001
1	B	01
2	C	1
3	D	000

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	001
1	B	01
2	C	1
3	D	000

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

C	0.40	1		
B	0.27	0	1	
A	0.20		0	1
D	0.13			0

0	A	001
1	B	01
2	C	1
3	D	000

0.40	2
0.27	1
0.20	0
0.13	3

Algoritmo de Shannon-Fano

A	0.20	001
B	0.27	01
C	0.40	1
D	0.13	000

$$H(S) = 1.89 \text{ bits}$$

$$L = 1.93 \text{ bits}$$

$$\eta = 0.9772$$

$$R = 0.0228$$

Algoritmo de Shannon-Fano

C A D B C C B A C A B C C D B

1 001 000 01 1 1 01 001 1 001 01 1 1 000 01

A	0.20	001
B	0.27	01
C	0.40	1
D	0.13	000

Implementación en Python

Algoritmo de Huffman

```
items = [[p, [i]] for i, p in enumerate(probs)]
```

Algoritmo de Shannon-Fano

```
items = [[p, i] for i, p in enumerate(probs)]
```

Codificación de fuentes de información

Teoría de la Información y la Comunicación

